

112.3%

IPSS
1.021 ms/step

What customers get

Static parser support for partial-phase feeders, source Thevenin impedance, linecode units, Kron reduction, transformer no-load and magnetizing admittance, OpenDSS load models, PV, storage, regulators, and capacitors.

Static power flow and balanced and unbalanced optimal power flow share three-phase APIs, facilitating model management and validation for planning and operational studies.

Time-series state application, enabled static controls, regulator and capacitor metadata, PV/storage profiles, and measured 8760-step performance.

IEEE 8500

0.001

pu max difference vs DSS-Python in IEEE8500 power-flow parity

8.9 s

**Total time for
8760-step QSTS
on a 7,522-node
feeder**

IPSS
8.9 s

DSS-
Python
10.0

Throughput
112.3%

Find feeder headroom for DER growth before queues become upgrades.

IPSS turns validated static feeders into repeatable screens for PV, storage, EV charging, electrification, and flexible load.

Voltage, thermal, unbalance, regulator, capacitor, and phase-specific constraints.

Location, size, profile, control mode, and seasonal loading sensitivities.

Balanced and unbalanced OPF identifies feasible headroom and upgrade alternatives.

8760-step QSTS checks controls and time-varying PV/storage behavior.

PV

Storage

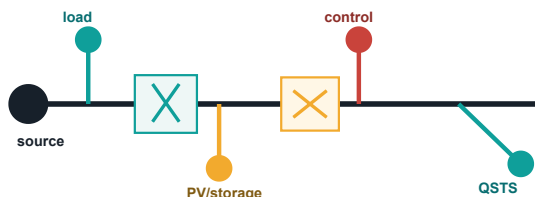
EV load

Flexible demand

Controls enabled

IPSS vs. DSS-Python: customer-relevant evidence

EVIDENCE POINT	IPSS	DSS-PYTHON REFERENCE	CUSTOMER VALUE
Controlled Ckt24 annual QSTS	1.021 ms/step, 8.948 s total, converged	1.147 ms/step, 10.046 s total, converged	Faster validated annual simulation
IEEE8500 PV-duty voltage parity	Max V delta 0.001047 pu; max angle delta 0.026743 deg	Controlled DSS-Python one-step export	Large-feeder voltage confidence
IEEE8500 branch-flow parity	Max branch P/Q delta 2.874 kW / 2.729 kvar; zero failures	Controlled DSS-Python terminal power export	Operational flow comparison
Static parser and PF gate	337 tests, 0 failures, 0 errors	DSS-Python mini-case and feeder references	Regression-backed delivery



- **Model depth:** single-, two-, and three-phase devices with active-phase handling for realistic distribution feeders.
- **Planning value:** one static model supports power flow, hosting and DER optimization workflows, and annual time-series studies.
- **Validation discipline:** comparisons are grounded in DSS-Python voltages, branch flows, load powers, generator powers, and capacitor states.